

Customer Shopping Behaviour Analysis Report

1. Project Overview

This project analyzes customer shopping behavior using Python (Pandas, NumPy) for data cleaning and feature engineering, and SQL for business-oriented analysis. The goal is to extract actionable insights related to revenue, customer segments, discounts, subscriptions, and purchasing patterns.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18 (after cleaning: 19, including engineered features)
- **Data Types:**
 - Numerical: Age, Purchase Amount, Review Rating, Previous Purchases
 - Categorical: Gender, Category, Shipping Type, Subscription Status, etc.

Key Columns

- **Purchase_amount** – customer spend per transaction
 - **review_rating** – product feedback score
 - **previous_purchases** – customer loyalty indicator
 - **frequency_of_purchases** – how often customers buy
-

3. Data Cleaning & Feature Engineering

3.1 Missing Values

- **Issue:** 37 missing values in **review_rating**
- **Solution:** Filled using the **median review rating within each product category** (more accurate than global median).

3.2 Column Standardization

- Converted all column names to lowercase
- Replaced spaces with underscores for SQL compatibility

3.3 Age Group Segmentation

Customers were segmented into four groups using quartiles: - Young - Adult - Middle Age - Senior

3.4 Purchase Frequency (Days)

Mapped purchase frequency labels into numeric days: - Weekly → 7 days - Fortnightly / Bi-Weekly → 14 days - Monthly → 30 days - Quarterly → 90 days - Annually → 365 days

4. SQL Analysis & Business Insights

Q1. Revenue by Gender

- Calculated total revenue generated by male and female customers.
- **Insight:** Shows which gender contributes more to overall sales.

	gender	total
▶	Male	157890
	Female	75191

Q2. Discount Users Spending Above Average

- Identified customers who used discounts but still spent **above the average purchase amount**.
- **Insight:** Discounts do not always reduce revenue; some high-value customers use them strategically.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97

Q3. Top 5 Products by Average Review Rating

- Ranked products based on average customer ratings.
- **Insight:** These products represent high customer satisfaction and brand trust.

item_purchased	averageRating
Gloves	3.9
Sandals	3.8
Boots	3.8
Hat	3.8
Skirt	3.8

Q4. Shipping Type vs Average Spend

- Compared average purchase amounts between **Standard** and **Express** shipping.
- **Insight:** Faster shipping is often associated with higher-value purchases

shipping_type	avg(purchase_amount)
Express	60.4752
Standard	58.4602

Q5. Subscription Impact on Spending

- Compared subscribers vs non-subscribers by:
- Average spend
- Total revenue
- **Insight:** Subscribed customers contribute higher lifetime value.

subscription_status	count(subscription_status)	averageSpend	sum(purchase_amount)
No	2847	59.8651	170436
Yes	1053	59.4919	62645

Q6. Products with Highest Discount Usage (%)

- Calculated percentage of discounted purchases per product.
- **Insight:** Highlights products that are most promotion-driven.

item_purchased	discount_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Q7. Customer Segmentation by Loyalty

Customers were segmented as: - **New:** 1 previous purchase - **Returning:** 2–10 purchases - **Loyal:** More than 10 purchases

- **Insight:** Loyal customers form a strong base for long-term revenue.

customer_segment	No of customer
Loyal	3116
Returning	701
New	83

Q8. Repeat Buyers vs Subscription Status

- Analyzed customers with more than 5 previous purchases.
- **Insight:** Repeat buyers are more likely to subscribe, indicating trust and engagement.

subscription_status	count(customer_id)
Yes	958
No	2518

Q9. Revenue Contribution by Age Group

- Summed total revenue per age group.
- **Insight:** Identifies the most profitable age segment for targeted marketing.

age_group	total
young	62143
middle age	59197
adult	55978
senior	55763

5. Key Business Takeaways

- Loyalty and subscriptions strongly influence revenue.
- Discounts can coexist with high spending behavior.
- Certain age groups and shipping preferences drive higher sales.
- Product ratings are useful indicators for premium and promotional strategies.

6. Tools & Technologies Used

- **Python:** Pandas, NumPy, Matplotlib, Seaborn
- **Database:** MySQL
- **Skills Demonstrated:**
 - Data cleaning & transformation
 - Feature engineering
 - SQL analytics & business querying
 - Customer segmentation

7. Conclusion

This analysis demonstrates an end-to-end data workflow: from raw data cleaning in Python to insight-driven SQL queries. The project reflects real-world analytics tasks commonly expected from data analysts and AI/ML engineers working with customer behavior data.

Power BI Data Visualization

